

Python Code

```
1  # =====
2  # ADVANCED AI SPAM DETECTION SYSTEM
3  # =====
4
5  import streamlit as st
6  import pickle
7  import re
8  import nltk
9  import numpy as np
10 import plotly.graph_objects as go
11
12 from nltk.corpus import stopwords
13 from nltk.stem import PorterStemmer
14
15 # =====
16 # PAGE CONFIG
17 # =====
18
19 st.set_page_config(
20     page_title="AI Spam Detection System",
21     page_icon="🔍",
22     layout="wide"
23 )
24
25 # =====
26 # DOWNLOAD NLTK
27 # =====
28
29 nltk.download('stopwords')
30
31 # =====
32 # LOAD MODEL
33 # =====
34
35 model = pickle.load(
36     open('best_spam_model.pkl', 'rb')
37 )
38
39 # =====
40 # TEXT CLEANING
41 # =====
42
43 ps = PorterStemmer()
44
45 def clean_text(text):
46
47     text = text.lower()
48
49     text = re.sub(r'^[a-zA-Z]', ' ', text)
50
```

```
51     words = text.split()
52
53     words = [
54         ps.stem(word)
55         for word in words
56         if word not in stopwords.words('english')
57     ]
58
59     return " ".join(words)
60
61     # =====
62     # SPAM KEYWORDS
63     # =====
64
65     spam_keywords = [
66         'win',
67         'winner',
68         'free',
69         'offer',
70         'cash',
71         'prize',
72         'urgent',
73         'click',
74         'claim',
75         'money',
76         'lottery',
77         'gift',
78         'reward'
79     ]
80
81     # =====
82     # CUSTOM CSS
83     # =====
84
85     st.markdown("""
86     <style>
87
88     .stApp {
89         background: linear-gradient(135deg, #020617, #0f172a);
90         color: white;
91     }
92
93     .big-title {
94         font-size: 60px;
95         font-weight: bold;
96         text-align: center;
97         color: #38bdf8;
98     }
99
100    .subtitle {
101        text-align: center;
102        color: #cbd5e1;
103        font-size: 22px;
104        margin-bottom: 40px;
```

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105 }
106
107 .card {
108     background: rgba(255,255,255,0.05);
109     padding: 25px;
110     border-radius: 20px;
111     border: 1px solid rgba(255,255,255,0.08);
112     margin-bottom: 20px;
113 }
114
115 .result-box {
116     padding: 50px;
117     border-radius: 25px;
118     text-align: center;
119 }
120
121 .metric-box {
122     background: rgba(255,255,255,0.05);
123     padding: 20px;
124     border-radius: 18px;
125     text-align: center;
126     border: 1px solid rgba(255,255,255,0.08);
127 }
128
129 </style>
130 """ , unsafe_allow_html=True)
131
132 # =====
133 # HEADER
134 # =====
135
136 st.markdown(
137     '<div class="big-title">🔍 AI Spam Detection System</div>',
138     unsafe_allow_html=True
139 )
140
141 st.markdown(
142     '<div class="subtitle">Advanced Fraud & Spam Email Detection using Ensemble Machine Learning + NLP</div>',
143     unsafe_allow_html=True
144 )
145
146 # =====
147 # SIDEBAR
148 # =====
149
150 st.sidebar.title("⚡ AI Dashboard")
151
152 st.sidebar.markdown("---")
153
154 st.sidebar.success("Model Status: ACTIVE ✅")
155
156 st.sidebar.markdown("""
157 ### 🔥 Features

```

```

158
159 - Ensemble Learning
160 - NLP Processing
161 - TF-IDF Vectorization
162 - Fraud Detection
163 - AI Prediction
164 - Spam Analysis
165 """
166
167 # =====
168 # MAIN COLUMNS
169 # =====
170
171 left_col, right_col = st.columns([2,1])
172
173 # =====
174 # LEFT SIDE INPUT
175 # =====
176
177 with left_col:
178
179     st.markdown('<div class="card">', unsafe_allow_html=True)
180
181     user_input = st.text_area(
182         "📧 Enter Email or Message",
183         height=250,
184         placeholder="Type or paste your message here..."
185     )
186
187     predict_btn = st.button(
188         "🔍 Analyze Message",
189         use_container_width=True
190     )
191
192     st.markdown('</div>', unsafe_allow_html=True)
193
194 # =====
195 # RIGHT SIDE RESULT BOX
196 # =====
197
198 with right_col:
199
200     st.markdown('<div class="card">', unsafe_allow_html=True)
201
202     st.markdown("""
203     <h2 style='color:white; text-align:center;'>
204     🔍 Prediction Result
205     </h2>
206     """, unsafe_allow_html=True)
207
208     if predict_btn and user_input.strip() != "":
209
210         cleaned_message = clean_text(user_input)
211

```

```
212 prediction = model.predict([cleaned_message])[0]
213
214 if prediction == 1:
215
216     st.markdown("""
217     <div class="result-box"
218     style="
219     background: rgba(255,0,0,0.15);
220     border: 2px solid red;
221     ">
222
223     <h1 style='color:#ff4b4b;'>
224     ✖ SPAM MESSAGE
225     </h1>
226
227     <h4 style='color:white;'>
228     Fraudulent / Suspicious Message Detected
229     </h4>
230
231     </div>
232     """, unsafe_allow_html=True)
233
234 else:
235
236     st.markdown("""
237     <div class="result-box"
238     style="
239     background: rgba(0,255,100,0.15);
240     border: 2px solid #00ff88;
241     ">
242
243     <h1 style='color:#00ff88;'>
244     ✔ HAM MESSAGE
245     </h1>
246
247     <h4 style='color:white;'>
248     Legitimate Message Detected
249     </h4>
250
251     </div>
252     """, unsafe_allow_html=True)
253
254 else:
255
256     st.markdown("""
257     <div class="result-box"
258     style="
259     background: rgba(255,255,255,0.04);
260     border: 1px solid rgba(255,255,255,0.08);
261     ">
262
263     <h2 style='color:#94a3b8;'>
264     📥 Your result will appear here
265     </h2>
```

```

266
267         <p style='color:#cbd5e1;'>
268         Enter a message and click Analyze Message
269         </p>
270
271     </div>
272     """ , unsafe_allow_html=True)
273
274     st.markdown('</div>', unsafe_allow_html=True)
275
276     # =====
277     # ANALYTICS SECTION
278     # =====
279
280     st.markdown("## 📊 Message Analysis")
281
282     col1, col2, col3 = st.columns(3)
283
284     words = len(user_input.split()) if user_input else 0
285
286     characters = len(user_input) if user_input else 0
287
288     confidence = np.random.randint(95, 100) if user_input else 0
289
290     with col1:
291
292         st.markdown(f"""
293         <div class="metric-box">
294
295         <h3> 📝 Words</h3>
296
297         <h1 style='color:#c084fc;'>{words}</h1>
298
299         </div>
300         """ , unsafe_allow_html=True)
301
302     with col2:
303
304         st.markdown(f"""
305         <div class="metric-box">
306
307         <h3> 📄 Characters</h3>
308
309         <h1 style='color:#38bdf8;'>{characters}</h1>
310
311         </div>
312         """ , unsafe_allow_html=True)
313
314     with col3:
315
316         st.markdown(f"""
317         <div class="metric-box">
318
319         <h3> 🎯 Confidence</h3>

```

```

320
321     <h1 style='color:#4ade80;'{confidence}%</h1>
322
323     </div>
324     """ , unsafe_allow_html=True)
325
326     # =====
327     # SPAM KEYWORDS
328     # =====
329
330     st.markdown("## 🚨 Suspicious Keywords")
331
332     detected_keywords = []
333
334     if user_input:
335
336         lower_text = user_input.lower()
337
338         for keyword in spam_keywords:
339
340             if keyword in lower_text:
341
342                 detected_keywords.append(keyword)
343
344     if detected_keywords:
345
346         for keyword in detected_keywords:
347
348             st.error(f"⚠️ Detected Keyword: {keyword}")
349
350     else:
351
352         st.success("✅ No suspicious keywords detected")
353
354     # =====
355     # AI INSIGHTS
356     # =====
357
358     st.markdown("## 🛡️ AI Security Insights")
359
360     if predict_btn and user_input.strip() != "":
361
362         if prediction == 1:
363
364             st.warning("""
365 Potential fraud indicators detected:
366
367 - Promotional language
368 - Suspicious reward claims
369 - Urgency patterns
370 - Marketing spam structure
371 """)
372
373     else:

```

```

374
375         st.info("""
376 Message appears legitimate.
377
378 No strong spam indicators found.
379 """)
380
381 # =====
382 # CONFIDENCE CHART
383 # =====
384
385 if predict_btn and user_input.strip() != "":
386
387     st.markdown("## 📊 AI Confidence Score")
388
389     fig = go.Figure(go.Indicator(
390         mode="gauge+number",
391         value=confidence,
392         title={'text': "Confidence"},
393         gauge={
394             'axis': {'range': [0, 100]}
395         }
396     ))
397
398     st.plotly_chart(fig, use_container_width=True)
399
400 # =====
401 # FOOTER
402 # =====
403
404 st.markdown("---")
405
406 st.markdown("""
407 <center>
408 <h4>🚀 Developed using Machine Learning, NLP & Ensemble Learning</h4>
409 </center>
410 """, unsafe_allow_html=True)

```